# Data_Scientist — Step 7

## Apply for Data Scientist Roles

Detailed Learning Notes • CareerCompass

## 1. Learning Objective

Prepare a portfolio, resume and interview strategy that demonstrates statistical thinking, coding, ML and business communication.

## 2. Core Concepts — Detailed Breakdown

### Portfolio

Show 2–4 projects with different strengths: analysis, experimentation, predictive modeling and end-to-end delivery. Each should explain the question, data, method, result and limitations.

### Technical interview

Prepare Python, SQL, statistics, probability, ML algorithms, metrics and data manipulation.

### Case interviews

Practice clarifying ambiguous questions, defining metrics, identifying data sources, proposing analysis and communicating uncertainty.

### Behavioral communication

Use a concise problem-action-result structure and be ready to discuss failures and trade-offs.

### Resume and applications

Use evidence-based bullets and match skills to the job description without claiming technologies you have not used.

## 3. Recommended Learning Workflow

- Polish two strongest projects.
- Practice SQL and Python every week.
- Practice one statistics/ML case per session.
- Prepare a two-minute project story.
- Tailor resume and application materials.
- Track applications and interview feedback.

## 5. Hands-on Project / Practice

Create a Data Scientist portfolio landing page linking three repositories. For each project provide a 30-second, 2-minute and 10-minute explanation so you can adapt to different interview formats.

## 6. Common Mistakes & Pitfalls

- Only listing certificates.
- Projects with no README or reproducibility.
- Memorizing interview answers without understanding.

- Ignoring SQL and statistics.
- Overstating model performance or business impact.

## 7. Completion Checklist

- I can explain my projects clearly.
- I can solve intermediate SQL/Python tasks.
- I can reason through an ML/statistics case.
- I can discuss limitations and trade-offs.
- My resume reflects demonstrated skills.

## 8. Interview / Assessment Questions

- Walk through your strongest analysis.
- How would you investigate a sudden metric drop?
- How do you choose an ML metric?
- Tell me about a failed experiment.
- How would you communicate uncertainty to a stakeholder?

## 9. Recommended References

Data Scientist Tutorial: https://www.youtube.com/watch?v=ua-CiDNNj30

Data Analyst Bootcamp: https://www.youtube.com/watch?v=PSNXoAs2FtQ

Kaggle Learn: https://www.kaggle.com/learn

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.